

Manjusha K

Research Analyst · Literature Review · Fact
Verification

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PROFILE

STEM research analyst with a B.Sc. in Statistics & Mathematics and an M.Sc. in Big Data Analytics (Generative AI specialization). Experienced in researching, verifying, and documenting technical claims against primary sources — statistical repositories, regulatory/clinical databases, and technical literature. Hands-on background building and testing LLM/RAG systems provides direct, practical insight into how AI models reason and where they fail, informing rigorous, source-verified evaluation of AI-generated content.

CORE STRENGTHS FOR RESEARCH & AI EVALUATION

Research & Verification: Multi-source fact-checking, technical literature review, regulatory/clinical data interpretation, structured documentation, audit trails

Statistical & Quantitative: Hypothesis testing (ANOVA, two-proportion z-test), time series forecasting, A/B testing, Excel/Power BI/Tableau

AI/LLM Systems Knowledge: LangChain, LlamaIndex, RAG pipelines, prompt engineering, LLM APIs (GPT, Gemini, Groq, Cohere), vector databases (Pinecone), multi-agent systems

Programming: Python (Pandas, NumPy, scikit-learn), R, SQL

Writing & Communication: Technical writing, data storytelling, cross-functional documentation

EXPERIENCE

AI/GenAI Engineer Intern — Trigent Software Pvt Ltd

Jan 2026 - Mar 2026 · Bengaluru

- Validated and debugged 800+ enterprise automation workflows against source specifications, systematically documenting discrepancies to improve reliability — direct, high-volume experience in structured verification and audit-standard documentation.
- Researched and cross-referenced clinical, pharmacological, and regulatory source material — drug-interaction data, renal dosing guidelines, HPO symptom-disease mappings — from technical literature to ground AI-driven healthcare tools in verified information.
- Built and stress-tested production RAG pipelines (LangChain, LlamaIndex, Pinecone) that retrieve and ground LLM responses in source documents, developing a working understanding of where retrieval-augmented systems succeed or fail at factual grounding.
- Designed and empirically validated an ML fraud-detection model (Random Forest, Logistic Regression) against transaction data prior to reporting accuracy claims (95%+), reinforcing a habit of verifying model outputs rather than taking them at face value.

RESEARCH PROJECTS

Handwriting-Based Parkinson's Disease Detection — Tools: Python, TensorFlow, scikit-learn, ResNet50, SVM

- Built and validated a hybrid deep-learning classifier using stratified 5-fold cross-validation, reporting 97.47% accuracy and 0.989 AUC — rigorous methodology and honest performance reporting over a single-split shortcut.

EV Adoption & Air Quality — Electrifying Bengaluru — Tools: Python, Pandas, NumPy, Statsmodels

- Analyzed 6 years (2018–2024) of public EV-adoption and air-quality data across multiple government sources to quantify the impact of policy interventions and forecast future PM2.5 trends under different scenarios.

A/B Testing — E-Commerce Conversion Optimization — Tools: Python, Pandas, Statsmodels

- Designed and executed a two-proportion z-test to verify a landing-page redesign's impact on conversion rate before it informed a product decision, rather than relying on raw before/after comparison.

EDUCATION

M.Sc. Big Data Analytics (Generative AI specialization)

St Joseph's University, Bengaluru · SGPA: 8.06

Jul 2024 - Present

B.Sc. Statistics & Mathematics

Mount Carmel College, Bengaluru · SGPA: 6.91

Aug 2021 - Apr 2024

CERTIFICATIONS

Google Analytics Certification — Great Learning

Goldman Sachs Excel Skills for Business — Forage

LEADERSHIP & ACTIVITIES

Head of Media & Marketing Committee — Datagram 7.0, St. Joseph's University

Core Team Member — Reel Run Association & Mathematics Association, Mount Carmel College

Student Intern — Under25 Summit 2023